

Bitwise differential trail through **ARADI** found by SAT

CiVerLy

July 7, 2026

1 ARADI

Maximal differential probability: 2^{-50}

0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	8	0	0	4	0	0	0	0	8	4	9	4	0	0	c	2	9
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	8	0	0	4	0	0	0	0	8	4	9	4	0	0	c	2	9
ARADI ← round0																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	4	0	1	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	4	0	1	
ARADI ← round1																																
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
ARADI ← round2																																
4	0	0	0	0	2	0	1	4	0	0	0	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	2	0	1	4	0	0	0	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ARADI ← round3																																
4	0	c	0	0	1	8	1	0	0	0	1	0	2	0	2	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	0	0
4	0	c	0	0	1	8	1	0	0	0	1	0	2	0	2	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	0	0
ARADI ← round4																																
0	2	a	6	c	9	c	d	0	0	0	0	0	0	0	0	8	8	8	1	1	3	1	2	0	0	0	0	0	0	0	0	0
0	2	a	6	c	9	c	d	0	0	0	0	0	0	0	0	8	8	8	1	1	3	1	2	0	0	0	0	0	0	0	0	0
PostRK																																
0	2	a	6	c	9	c	d	0	0	0	0	0	0	0	0	8	8	8	1	1	3	1	2	0	0	0	0	0	0	0	0	0
0	2	a	6	c	9	c	d	0	0	0	0	0	0	0	0	8	8	8	1	1	3	1	2	0	0	0	0	0	0	0	0	0

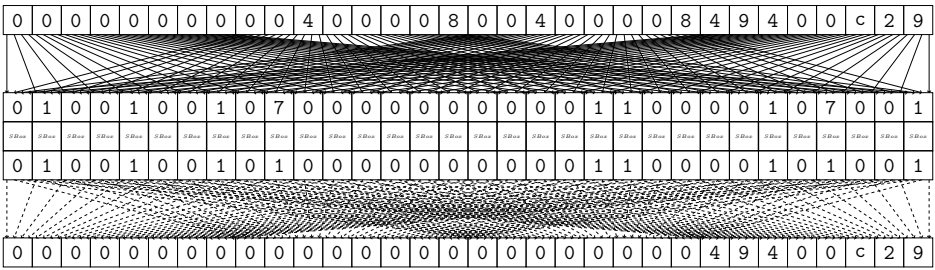
2 ARADI-round0

Maximal differential probability: $2^{-18.0}$

0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	8	0	0	4	0	0	0	0	8	4	9	4	0	0	c	2	9
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	8	0	0	4	0	0	0	0	8	4	9	4	0	0	c	2	9
RR																																
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	8	0	0	4	0	0	0	0	8	4	9	4	0	0	c	2	9
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	8	0	0	4	0	0	0	0	8	4	9	4	0	0	c	2	9
SBox Layer																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	9	4	0	0	c	2	9
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	9	4	0	0	c	2	9
Linear Layer0																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	4	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	4	0	1

3 SBoxLayer

Maximal differential probability: $2^{-18.0}$



4 LinearLayer0

Maximal differential probability: $2^{-0.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	9	4	0	0	c	2	9
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	9	4	0	0	c	2	9	
xor								xor								xor								xor							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	4	0	1		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	4	0	1		

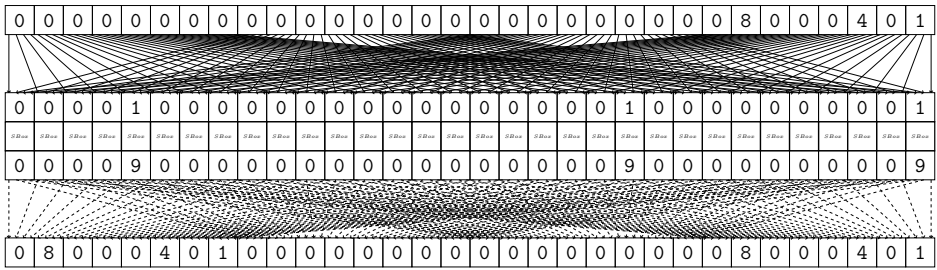
5 ARADI-round1

Maximal differential probability: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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6 SBoxLayer

Maximal differential probability: $2^{-6.0}$



7 LinearLayer1

Maximal differential probability: $2^{-0.0}$

0	8	0	0	0	4	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	4	0	1	
0	8	0	0	0	4	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	4	0	1	
x0								x1								x2								x3											
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

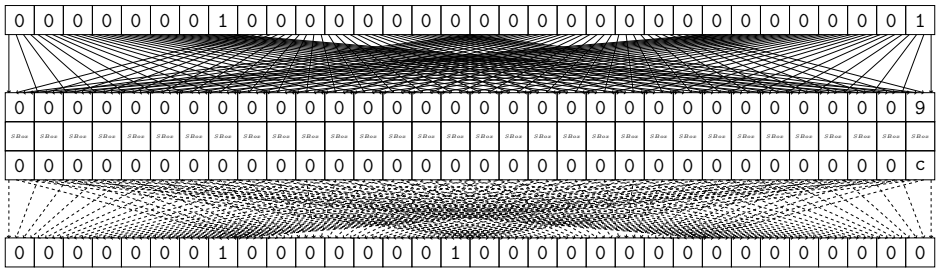
8 ARADI-round2

Maximal differential probability: $2^{-2.0}$

0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
RR																											
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
3 Bias Layer																											
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	
Linear Layer2																											
4	0	0	0	0	2	0	1	4	0	0	0	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	
4	0	0	0	0	2	0	1	4	0	0	0	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	

9 SBoxLayer

Maximal differential probability: $2^{-2.0}$



10 LinearLayer2

Maximal differential probability: $2^{-0.0}$ [illegible]

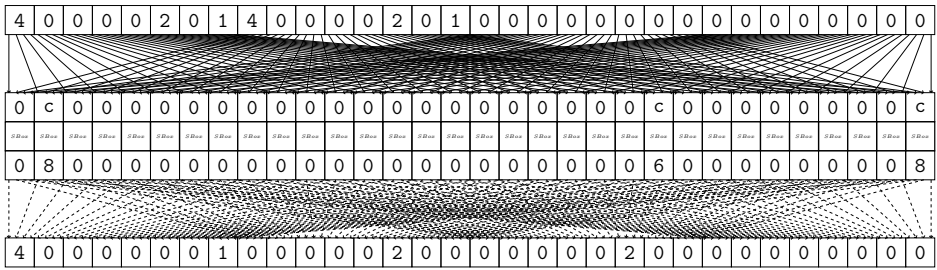
11 ARADI-round3

Maximal differential probability: $2^{-6.0}$

[illegible]

12 SBoxLayer

Maximal differential probability: $2^{-6.0}$



13 LinearLayer3

Maximal differential probability: $2^{-0.0}$ [illegible]

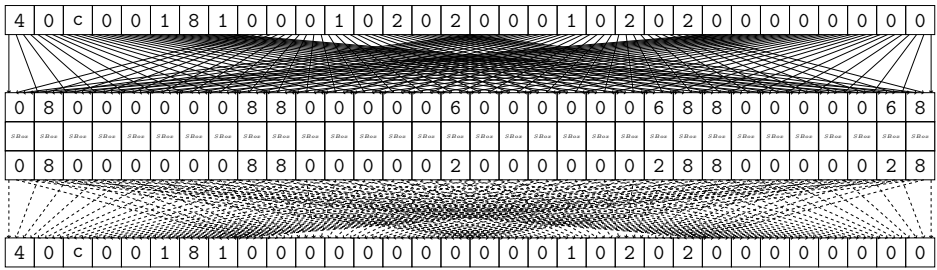
14 ARADI-round0

Maximal differential probability: $2^{-18.0}$

4	0	c	0	0	1	8	1	0	0	0	1	0	2	0	2	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	0
4	0	c	0	0	1	8	1	0	0	0	1	0	2	0	2	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	0
4	0	c	0	0	1	8	1	0	0	0	1	0	2	0	2	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	0
4	0	c	0	0	1	8	1	0	0	0	1	0	2	0	2	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	0
4	0	c	0	0	1	8	1	0	0	0	0	0	0	0	0	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	0
4	0	c	0	0	1	8	1	0	0	0	0	0	0	0	0	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	0
0	2	a	6	c	9	c	d	0	0	0	0	0	0	0	0	8	8	8	1	1	3	1	2	0	0	0	0	0	0	0	0
0	2	a	6	c	9	c	d	0	0	0	0	0	0	0	0	8	8	8	1	1	3	1	2	0	0	0	0	0	0	0	0

15 SBoxLayer

Maximal differential probability: $2^{-18.0}$



16 LinearLayer0

Maximal differential probability: $2^{-0.0}$

4	0	c	0	0	1	8	1	0	0	0	0	0	0	0	0	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	
4	0	c	0	0	1	8	1	0	0	0	0	0	0	0	0	0	0	0	1	0	2	0	2	0	0	0	0	0	0	0	
xor								xor								xor								xor							
0	2	a	6	c	9	c	d	0	0	0	0	0	0	0	0	8	8	8	1	1	3	1	2	0	0	0	0	0	0	0	
0	2	a	6	c	9	c	d	0	0	0	0	0	0	0	0	8	8	8	1	1	3	1	2	0	0	0	0	0	0	0	